PROJECT REPORT

Design and Implementation of a Database Management System for Pharmacy Prescription and Medicine Stock Management System

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2. Abstract

This project uses MySQL database pharmacy_management to organize medicines, patients, and prescriptions. The sample dataset contains three medicines with quantities 100, 50, and 75, three patients, and no prescription rows. The Presentation-II query for quantity = 0 returns no rows. Real UI screenshots were not available in saved assets, so labelled slots are provided.

3. Introduction and problem statement

Relational tables provide structured storage for medicine stock, patients, and prescriptions. The project aims to reduce disorganized records and enable stock queries. Unverified features are not claimed.

4. Objectives and scope

Create the database; maintain medicine and patient records; connect prescriptions; run retrieval and zero-stock queries. Billing, supplier purchasing, authentication, expiry alerts, and automatic stock deduction are not verified.

5. Software and hardware requirements

Type	Requirement
Software	MySQL Server, MySQL Workbench, supported OS; UI runtime if applicable.

Hardware	Laptop/desktop; recommended 4 GB RAM or more and free disk space.
Connectivity	Configured MySQL host, port, username, password.

6. ER diagram

The ER diagram shows PATIENTS, PRESCRIPTIONS, and MEDICINES, with prescription records linking patients and medicines.

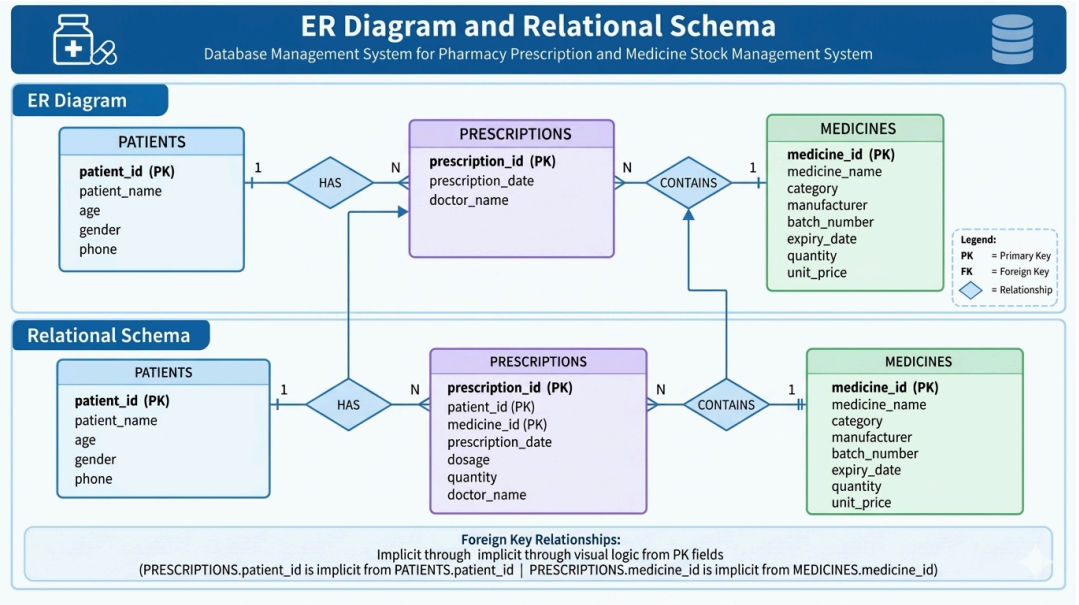


Figure 1. Saved ER diagram and relational schema.

7. Relational schema and normalization

Relation	Purpose
PATIENTS	Patient details; patient_id identifier.
MEDICINES	Medicine details and quantity; medicine_id identifier.
PRESCRIPTIONS	Associates patient and medicine IDs.

Primary keys identify rows. Foreign keys should enforce valid references if declared. Confirm normalization from exact functional dependencies.

8. Data dictionary

Table	Field	Type / constraint guidance
PATIENTS	patient_id	INT primary key
PATIENTS	name, gender, phone	VARCHAR; required fields NOT NULL
PATIENTS	age	INT
MEDICINES	medicine_id	INT primary key
MEDICINES	medicine_name, category	VARCHAR
MEDICINES	quantity	INT; non-negative
MEDICINES	price if present	DECIMAL(10,2)

PRESCRIPTIONS	prescription_id	INT primary key
PRESCRIPTIONS	patient_id, medicine_id	INT foreign keys if declared
PRESCRIPTIONS	date/dosage/duration	Types per actual SQL

9. SQL commands used (DDL, DML) with sample outputs

```
CREATE DATABASE pharmacy_management;
USE pharmacy_management;
SELECT * FROM medicines;
SELECT * FROM patients;
SELECT * FROM prescriptions;
```

Command	Purpose	Typical output
CREATE DATABASE / CREATE TABLE	Create database and tables.	Query OK if successful.
INSERT	Add records.	1 row affected.
SELECT	Read records.	Result grid.
UPDATE / DELETE	Modify/remove records.	Rows matched/affected.

10. Queries with outputs, including Presentation-II query

```
SELECT * FROM medicines WHERE quantity = 0;
```

Check	Output
Medicine table	3 rows; quantities 100, 50, 75.
Patient table	3 rows.
Prescription table	0 rows in approved dataset.
quantity = 0	0 rows returned.

11. UI design and screenshots

Original UI screenshots were not found in saved assets. Replace each labelled slot with a genuine screenshot.

Dashboard
Insert actual home screen. [SCREENSHOT SLOT]

Medicine inventory
Insert list and quantities. [SCREENSHOT SLOT]

Patient management
Insert patient page/form. [SCREENSHOT SLOT]

Prescription management
Insert prescription page/form. [SCREENSHOT SLOT]

Query result
Insert Workbench result grid. [SCREENSHOT SLOT]

12. Implementation details

Component	**Details**
Database	MySQL; pharmacy_management.
Client	MySQL Workbench.
Connectivity	Verify actual host/port; localhost:3306 was previously mentioned.
UI technology	Enter actual framework; not confirmed from saved assets.
Key query	SELECT * FROM medicines WHERE quantity = 0;

13. Testing

ID	**Test**	**Result**
TC-01	Medicines	3 records; quantities 100, 50, 75.
TC-02	Patients	3 records.
TC-03	Prescriptions	0 records.
TC-04	Zero-stock filter	0 rows; passes sample test.
TC-05	Valid prescription insert	Requires live verification.
TC-06	Invalid foreign key	Depends on defined FK constraint.

14. Conclusion and future enhancements

The database structures medicine, patient, and prescription records and supports a zero-stock query. Future work may add expiry tracking, low-stock alerts, prescription validation, transactional stock updates, access control, and expanded testing. Add genuine UI and SQL screenshots before submission.

15. References

MySQL Reference Manual: https://dev.mysql.com/doc/
Saved ER diagram/schema image.
Course project materials.

16. Appendix: GitHub repository link

https://github.com/LALA-52/DBMS-Course-Project